Name		Date of Performance	
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TITLE: Identification of surface roughness using interferometer.

APPARATUS:

INSTRUMENT	**SPECIFICATION**
Optical flat	One side flat
Monochromatic light source	Sodium vapor light

INTRODUCTION :

Optical flat is a glass surface polished optically flat upto a flatness of 0.5 micron and used as a standard for measurement and comparison of flat surface of various component micrometer measuring faces and other measuring instrument surface by optical interface.

Principle of Interference:

As stated earlier light is a form of energy being propagated by electromagnetic waves which may be represented by a sine curve. When two rays of the same wavelength meet at some point, a mutual interference occurs and the nature of interference will depend upon the phases of two waves at their meeting point.

If the waves are in same phase they will reinforce each other and the resulting intensity will be the sum of two intensities. However, if two waves are out of phase, the resultant intensity (also amplitude) will be the difference of two. If both the waves having same amplitude are in phase, the resultant amplitude and hence the intensity becomes twice, and the result will be a bright spot. However, if these two waves (with same amplitude) are out of phase the resultant intensity will be zero and the result will be a dark spot.

The interference can occur only when two light rays are coherent, that is the two rays maintain their phase relationship for an appreciable length of time. This is possible only when the two rays are originated from the same point of the light source at the same time.

Optical Flat:

The simplest illustration of the interference is the use of official flat. It provides precision as well as accuracy in the measurement of flatness. Optical flats are cylindrical pieces 25 to 300 mm in dia. with a thickness of about 1/6th of the dia. They are made of transparent material such as quartz, glass, sapphire, etc. Optical flats made of quartz are more commonly used because of its hardness, low coefficient of expansion, resistance to corrosion and much longer useful life. One or both surface of optical flats may be highly polished. An arrow is made on the flat to indicate finished surface. For measuring flatness, in addition to optical flat, the monochromatic light source, emitting light of single wavelength is also required. The yellow orange light radiated by helium gas is most satisfactory for use with optical flat.

Sometimes the optical flats are coated with a thin film of titanium oxide. This reduces the loss of light due to reflection which makes the band more clear. The coating is so thin that it does not affect the position of the fringe band but such a flat requires greater care. For greater accuracy optical flat must be used in an area where the temperature is constant. It is advisable to use lint free paper for cleaning the optical flat and the surface of the part to be checked.

Optical flats are of two types:

Type A:

It has single flat working surface. It is used for testing flatness of precision measuring surfaces of flats, slip gauges, measuring tables ,etc.

Type B:

It has both working surfaces flat and parallel to each other. It is used for test measuring surfaces of micrometer, measuring anvils , meters and similar measuring devices for testing flatness and parallelism.

In each type there are two grades. Reference grade or grade **I** and working grade or grade **II**. The tolerance on flatness for Grade **I** is 0.05 micrometer and for Grade **II** is 0.10 micrometer, tolerance on parallelism (type B) is 0.15 micrometer and 0.20 micrometer respectively and tolerance on thickness (type B) is 0.20 micrometer and 0.30 micrometer respectively **(IS.5440-1969).**

The flats are marked clearly on the cylindrical surface with the grade and type and an identifying serial number together with the manufacturers trademarks. For example,

1) Grade **I** type A flat of diameter 250mm is designated by : Optical flat **IA25-IS**:5440.
2) Grade **II** type B flat of thickness 12.125 mm is designated by optical flat **II** B 12.125- **IS:5440**.

The grade I flat is identified by a black dot about 1mm in diameter on the cylindrical surface of the flat and away from other markings.

Optical flats are used to test the flatness of lapped surfaces such as gauge blocks, gauges, micrometer anvils , etc. When an optical flat is placed on work piece surface it will not form an intimate contact, but will be at slight inclination to the surface, forming at airwedge between the surfaces.

If optical flat is now illuminated by monochromatic source of light, interference fringes will be observed. These are produced by the interference of light rays reflected from the bottom face of optical flat and top face of the workpiece being tested through the layers of air.

Consider a ray of light incident at 'A' on optical flat placed over a work piece to be tested. It passes through the optical flat and its bottom face is divided into two components. One component of the incident ray gets reflected from bottom of the optical flat at 'B' in the direction BC and the other portion, transmitted through the layer of the entrapped air will be reflected by the top face of the workpiece at D in the direction DEF.

The paths travelled by both the reflected rays differ by an amount BDE i.e the second component of the ray lags behind the first by an amount equal to twice the air gap. Though both the components have the same wavelength and start, difference in their paths causes them to be either in phase or out of phase at C and F.

As explained earlier, if the path difference between the reflected rays is even multiple of half wavelength, they will be out of phase and dark band will be observed. If the path difference is odd multiple of half wavelength they will be in phase with each other and will be reinforced each other. So brightness will be observed., Depending upon the air gap between surfaces , we will get alternate dark and bright bands due to interference of light.

LIMITATIONS OF OPTICAL FLAT:

The optical flat suffers from the following limitations per precise work:

1) It is difficult to control the lay of the optical flat and thus orient the fringes to the best advantage.
2) The fringe pattern is not viewed from directly above the resulting obliquity can cause distortion and errors in viewing.

Fingertip test :- Thus, if by light particle the curve of fingers are brought closer it is convex surface and the level of that place must be lowered down to form a flat surface. If the numbers of fingers are reduced and the fingers move apart, it is concave surface.

CONCLUSION:

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TITLE: Measurement of surface roughness using surface roughness tester.

INTRODUCTION:

With more precise demands of modern engineering products, the control of surface texture together with dimensional accuracy has become more important. It has been investigated that the surface texture greatly influences the functioning of the machined parts. The properties such as appearance, corrosion resistance, wear resistance, fatigue resistance, Lubrication, initial tolerance, ability to hold pressure, load carrying capacity, noise reduction in case of gears are influenced by the surface texture.

ELEMENTS OF SURFACE TEXTURE:

The various elements of surface texture can be defined and explained with the help of fig. which shows a typical surface highly magnified.

Surface: The surface of apart is confined by the boundary which separates that part from another part, substance or space.

Actual surface: This refers to the surface of a part which is actually obtained after a manufacturing process.

Nominal surface: A nominal surface is a theoretical, geometrically perfect surface which does not exist in practice, but it is an average of the irregularities that are superimposed on it.

Profile: Profile is defined as the contour of any section through a surface.

Roughness: As already explained roughness refers to relatively finely spaced microgeometrical irregularities. it is also called as primary texture and constitutes third and fourth order irregularities.

Roughness height: This is rated as the arithmetical average deviation expressed in micro-meters normal to an imaginary centre line, running through the roughness profile.

Roughness width: Roughness width is the distance parallel to the normal surface between successive peaks or ridges that constitutes the predominant pattern of the roughness.

(a) **C.L.A METHOD:** In this method, the surface roughness is measured as the average deviation from the nominal surface.
centre line average or arithemetic average is defined as the average values of the ordinates from the mean line, regardless of the arithmetic signs of the ordinates.

$$\text{C.L.A. Value} = \frac{h_1 + h_2 + h_3 + \ldots\ldots\ldots h_n}{n}$$

And

$$\text{C.L.A. Value} = \frac{A_1 + A_2 + A_3 + \ldots\ldots\ldots A_n}{L}$$

$$\text{C.L.A. Value} = \frac{\sum A}{L}$$

The calculation of C.l.A. value using equation (ii) is facilitated by the planimeter.
CLA value measure is preferred to RMS value measure because its value can be easily determined by measuring the areas with planimeter or graph or can be readily determined in electrical instruments by intergraining the movement of the styles and displaying the result as an average.

(b)**R.M.S.METHOD:** In this method also, the roughness is measured as the average deviation from the nominal surface. Root mean square value measured is based on the least squares.
R.M.S values are defined as the square root of the arithmetic mean of the values of the squares of the ordinates of the surface measured from a mean line. It is obtained by setting many equidistant ordinates on the mean line (y1, y2, y3,.........yn) and then taking the root of the mean of the squared ordinates.
Let us assume that the sample length 'L' is divided into 'n' equal parts and y1, y2, y3, ... yn, are the height of the ordinates erected at those points.

Stylus which is a fine point made of diamond or any such hard material is drawn over the surface to be tested. The movements of the stylus are used to modulate a high frequency carrier current to generate a voltage signal. The output is then amplified by suitable means and used to operate a recording or indicating instrument.

Stylus type instrument generally consist of the following units:

1. Skid or shoe.
2. Finely pointed stylus or probe.
3. An amplifying device for magnifying the stylus movement and indicator.
4. Recording device to produce a trace and
5. Means for analyzing the trace.

Skid or shoe is drawn slowly over the surface either by hand or by motor drive. It follows the general contours of the surface and provides a datum for measurements. The stylus moves over the surface with the skid. It moves vertically up and down due to surface roughness and records the micro-geometrical form of the surface. The stylus movements are magnified by an amplifying device and recorded to produce a trace. The trace is then analyzed by some automatic device incorporated in the instrument.

OBSERVATION TABLE:

Sr. No.	Component	Ra Value
1	Slip Gauge	

Conclusion:

Question:

Q1) What are roughness comparison specimen? How they assess surface roughness.
Q2) It is not possible to produce perfectly smooth surface. Justify the statement.
Q3) State the reason for controlling the surface finish.